SAE & Causal Analysis of Malaria Interventions in West Africa

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Background

Malaria remains a major public health challenge in West Africa. Demographic Health Surveys (DHS) provide essential prevalence data but are designed for national-level precision[1]. At the district level, small sample sizes produce unreliable estimates[2], and area-level causal inference risks ecological fallacy.

Primary Aim: To develop reliable district-level estimates of malaria burden and intervention coverage using Small Area Estimation methods.

Sub Aims:

1. To Calculate survey-weighted direct estimates and assess reliability.
2. To Apply Fay-Herriot models to produce smoothed small area estimates.
3. To Compare independent vs. spatial random effects specifications.
4. To Investigate causal effects of interventions at small area level *(Future)*.

Methodology

Models were fitted using Bayesian inference in Stan[3]

Survey-Weighted Direct Estimation

$$\hat{p}_i = \frac{\sum w_j y_j}{\sum w_j}$$

where w_j = sampling weight, y_j = outcome for individual j in district i.

Fay-Herriot Model

Sampling: $\hat{\theta}_i | \theta_i \sim N(\theta_i, \psi_i^2)$

Linking (IID): $\theta_i = \mathbf{x}_i'\boldsymbol{\beta} + u_i$, where $u_i \sim N(0, \sigma_u^2)$

Linking (Spatial): $\theta_i = \mathbf{x}_i'\boldsymbol{\beta} + \phi_i$, where ϕ follows ICAR prior

Shrinkage Estimator:

$$\hat{\theta}_i^{SAE} = \gamma_i \hat{\theta}_i^{DIR} + (1 - \gamma_i)\mathbf{x}_i'\boldsymbol{\beta}$$

where $\gamma_i = \frac{\sigma_u^2}{\sigma_u^2 + \psi_i^2}$ balances data reliability vs. model prediction.

Data Sources

Source	Variable	Description
Ghana DHS 2022[4]	Malaria prev (Outcome)	Microscopy test (~2 wk)
	Artemisinin-based Combination Therapy (ACT)	ACT for fever
	Insecticide-Treated Net (ITNs) (Covariate)	Slept under net
	Survey design	Strata, clusters, weights
MAP	Plasmodium falciparum Parasite Rate (PfPR, Covariate)	Modeled parasite rate
	Accessibility (covariate)	Travel time to facility
WHO	Boundaries	District polygons

Study Population: Children 6-59 months, the WHO standard indicator population for malaria prevalence, serving as a proxy for national transmission intensity[5].

DHS surveys are conducted over approximately 3+ months. Seasonal variation in malaria transmission may affect estimates from regions measured at different times of year

Early Results

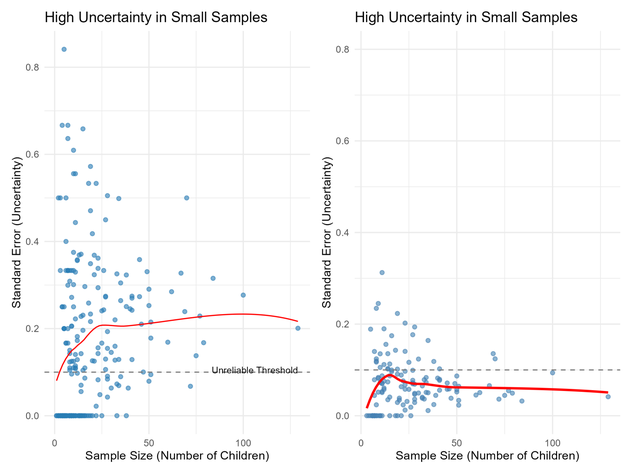

Figure 1: SE vs Sample Size ~ Direct and SAE Estimates (Malaria Prevelance)

As shown in Figure 1, direct survey estimates have high uncertainty in small-sample districts, with 70% showing CV > 20%.

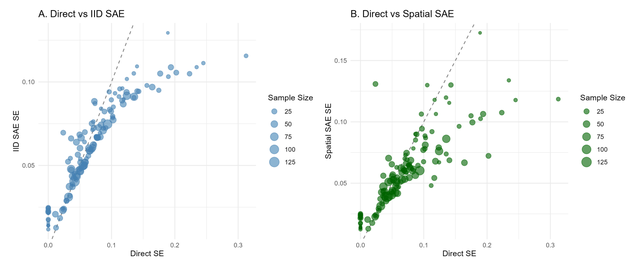

Figure 2: SE vs Sample Size (Independently and Identically Distributed (IID) & Spatial Intrinsic Conditional Autoregressive (ICAR) (Malaria Prevelance)

Small Area Estimation reduces standard errors by borrowing strength from model predictions and neighboring districts

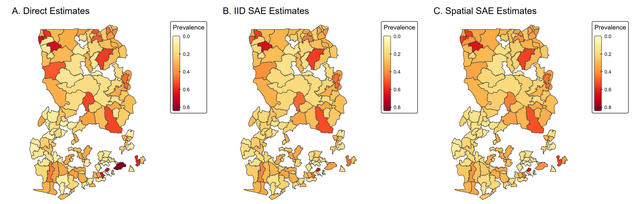

Figure 3: Ghana Malaria Prelalemce (Direct, IID, Spatial

Spatial SAE produces smoothest estimates while preserving true geographic variation. SE reduced by 17%, increasing reliable districts from 30% to 38%

Model	Sigma Parameter	Mean SE	SE Reduction	Reliable CV (<20%)
Direct Estimates	•	0.071	•	30% (41/137)
IID (no covariates)	sigma_mu = 0.66	0.060	14.5%	34% (47/137
IID + covairiates	sigma_mu = 0.66	0.060	14.5%	34% (47/137
Spatial (ICAR)	sigma phi = 0.99	0.059	17.0%	38% (52/137)

Model Selection (LOO-CV): Models were compared using Leave-One-Out Cross-Validation (LOO-CV), which estimates out-of-sample predictive accuracy.

Model	ELPD	SE	Ranking
Spatial ICAR	0.0 (ref)	•	Best
IID + Covariates	-2.3	1.8	2nd
IID (no covariates)	-2.5	1.9	3rd

The Spatial ICAR model was selected as it best captures the geographic clustering of malaria transmission. Notably, adding covariates (ITN, PfPR, Accessibility) did not improve model fit spatial structure explains more district-level variation than environmental predictors.

Next Project Steps

- Implement Besag-York-Mollié 2 model (BYM2 model)
- Add temporal smoothing with multi-year DHS
- Extend SAE to ITN/ACT coverage outcomes
- Apply to other West African countries
- Investigate the possibility of causal intervention effects based on SAE and develop Shiny dashboard for policy-makers

GitHub

The code and datasets for this project can be viewed at our GitHub repository here: https://github.com/SAHagen/HDS5106

References

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